

Lesson #6.00 2365 - Level 3 City and Guilds

Question example 3:

1

Question example 3:

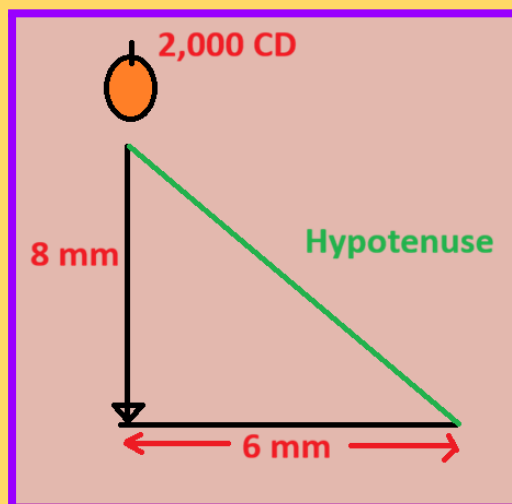
A street lighting column suspends a 2,000 cd light source. 8 metres above the ground. Determine the illuminance directly below the lamp and 6 metres to one side of the lamp base.

Pre-requisites

Cd -> how bright the light is.

D^2 -> distance to the ground

Optional - Draw a basic picture



Calculate the luminance directly under lamp first

$$\text{Luminance} = \frac{\text{cd}}{d^2} = \frac{2000}{8^2} = \frac{2000}{64} = 31.25$$

= 31.35 lux

Class date: 14/12/2025

Then, Calculate the luminance on the angled line

First, Calculate the hypotenuse.

$$c = \sqrt{a^2 + b^2}$$

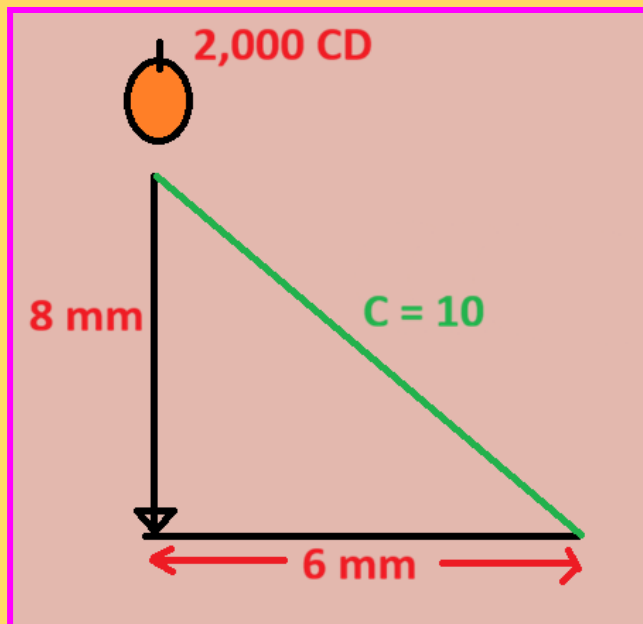
$$c = \sqrt{8^2 + 6^2}$$

$$c = \sqrt{64 + 36}$$

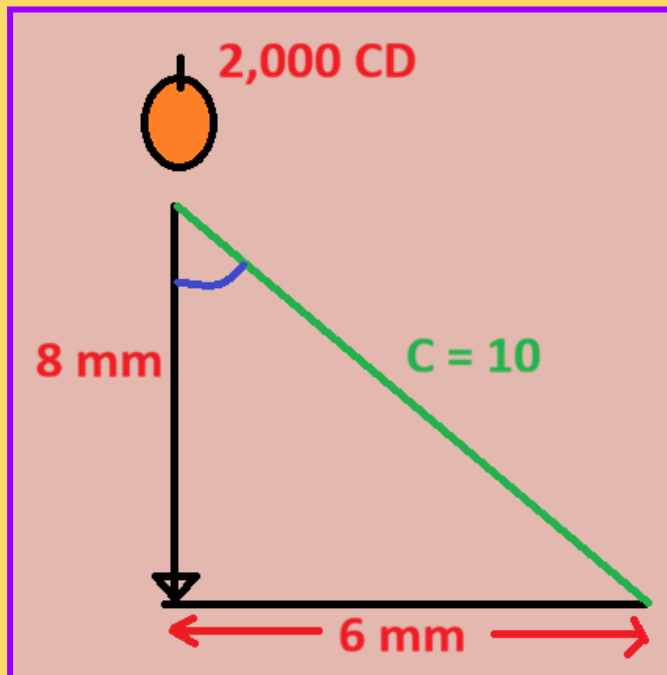
$$c = \sqrt{100}$$

$$c = 10$$

Update the picture with hypotenuse value



Next calculate the cosign.



The cosign is equal to the adjacent / hypotenuse.

Therefore, $8 / 10 \Rightarrow 0.8$.

Finale

$$\frac{cd}{d^2} \times \cos \theta$$

$$2,000 / 10^2 \times 0.8 = \mathbf{16 \text{ lux}}$$